ITC355: Assignment 2
Analyzing University Student Data for Retention Strategies Using Power BI

1. Problem Statement Recap:

Student dropout rates are a major challenge for universities which lead to financial losses, academic disruption & a damage to the institute's reputation. Identifying the key factors influencing retention and intervening on time is very important in order to improve the graduation rates.

This analysis report shall make use of the business intelligence analysis from Assignment 1 with the purpose of focusing on student dropouts at a university. This assignment will apply **descriptive** and **predictive analytics** to find patterns contributing to student attrition and to create a solution to improve retention.

Below is the dataset that will be used in this assignment:

churn_analysis.csv: This dataset contains information on student attendance, whether they receive financial aid, and if they dropped out.

Data Columns:

- **StudentID**: Unique identifier for each student.
- **Attendance**: Student attendance percentage.
- **FinancialAid**: Whether the student receives financial aid (1 = Yes, 0 = No).
- **Churn**: Whether the student dropped out (1 = Yes, 0 = No).

feature_importance.csv: This dataset includes GPA, attendance, and financial aid status, along with the churn (dropout) indicator.

Data Columns:

- **StudentID**: Unique identifier for each student.
- **GPA**: Student GPA.
- **Attendance**: Student attendance percentage.
- **FinancialAid**: Whether the student receives financial aid (1 = Yes, 0 = No).
- **Churn**: Whether the student dropped out (1 = Yes, 0 = No).

student_performance.csv: This dataset contains GPA, attendance, financial aid, and mental health information for each student.

Data Columns:

- **StudentID**: Unique identifier for each student.
- **GPA**: Student GPA.
- **Attendance**: Student attendance percentage.
- **FinancialAid**: Whether the student receives financial aid (1 = Yes, 0 = No).
- **MentalHealthStatus**: A numerical value representing the mental health status of the student.

retention_strategy.csv: This dataset outlines the tasks involved in the university's retention strategy, with timelines for each task to help prevent student dropout.

Data Columns:

- **Task**: The task in the retention strategy (e.g., "Provide Tutoring", "Offer Financial Aid").
- **Start**: The start date of the task (in YYYY-MM-DD format).
- **End**: The end date of the task (in YYYY-MM-DD format).

Finally, Microsoft Power BI Desktop shall be used in this assignment to create graphs & visuals in order to highlight factors of student dropouts & suggest data-driven interventions.

2. BI Analysis in Power BI Desktop:

A. Descriptive Analytics:

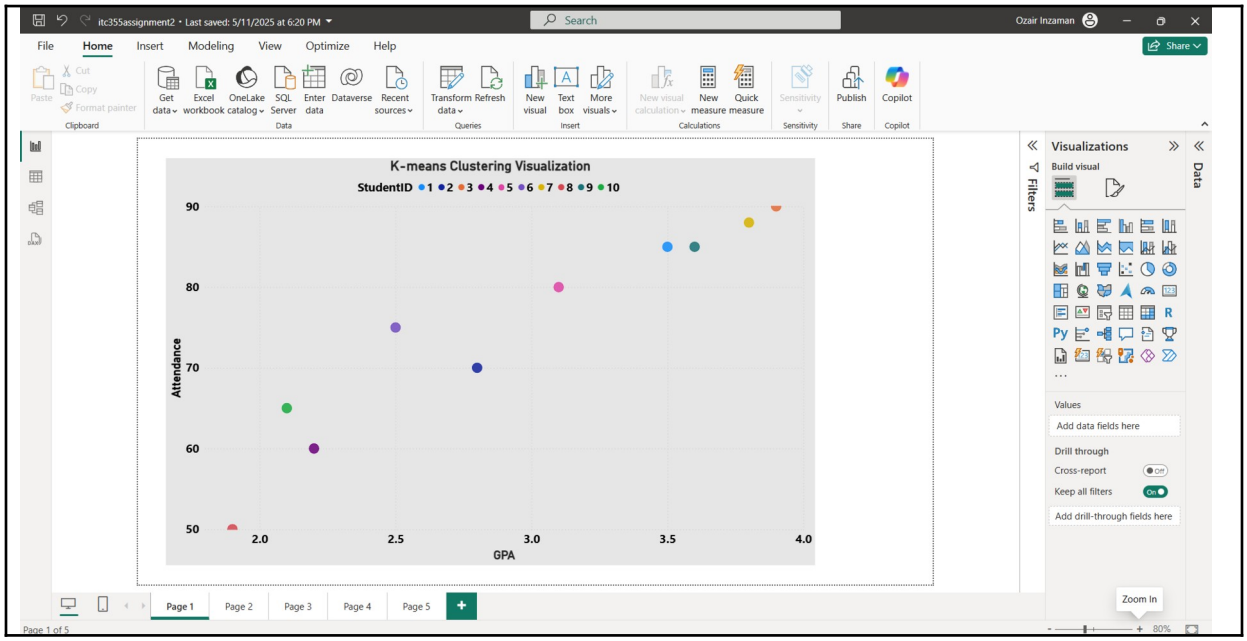
Descriptive Analytics was used to analyze historical data in order to understand the patterns and trends therefore revealing the various relationships between student behavior and academic outcomes.

The **GPA**, **attendance**, and **mental health status**, was used to identify patterns used to determine why students drop out or perform poorly. Clustering techniques were used considering **GPA** and **attendance** in order to identify students who are at high risk.

Clusters were created using **GPA** and **attendance** in Power BI’s Visualizations options.

The clusters highlighted the high-risk students for dropouts based on a **low GPA** and **attendance** therefore allowed focus to be made on the groups most likely to drop out and apply targeted interventions.

Graph 1: K-means Clustering of Students in Power BI

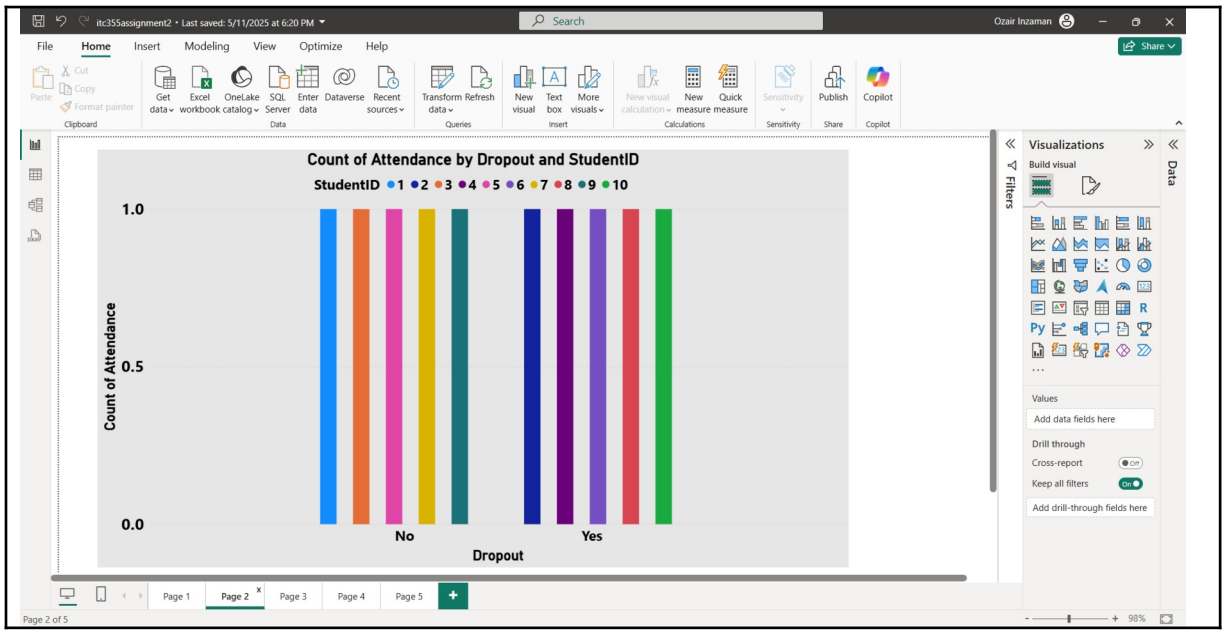


The chart above shows the amount of students that fall into each GPA range, therefore giving an idea if students with a low GPA are over-represented in the dropout category. The GPA is the main indicator of academic performance. Low GPAs are clearly related to high dropout risks. The chart also shows the distribution of GPAs across students thus allowing to identify students needing academic support.

B. Predictive Analytics:

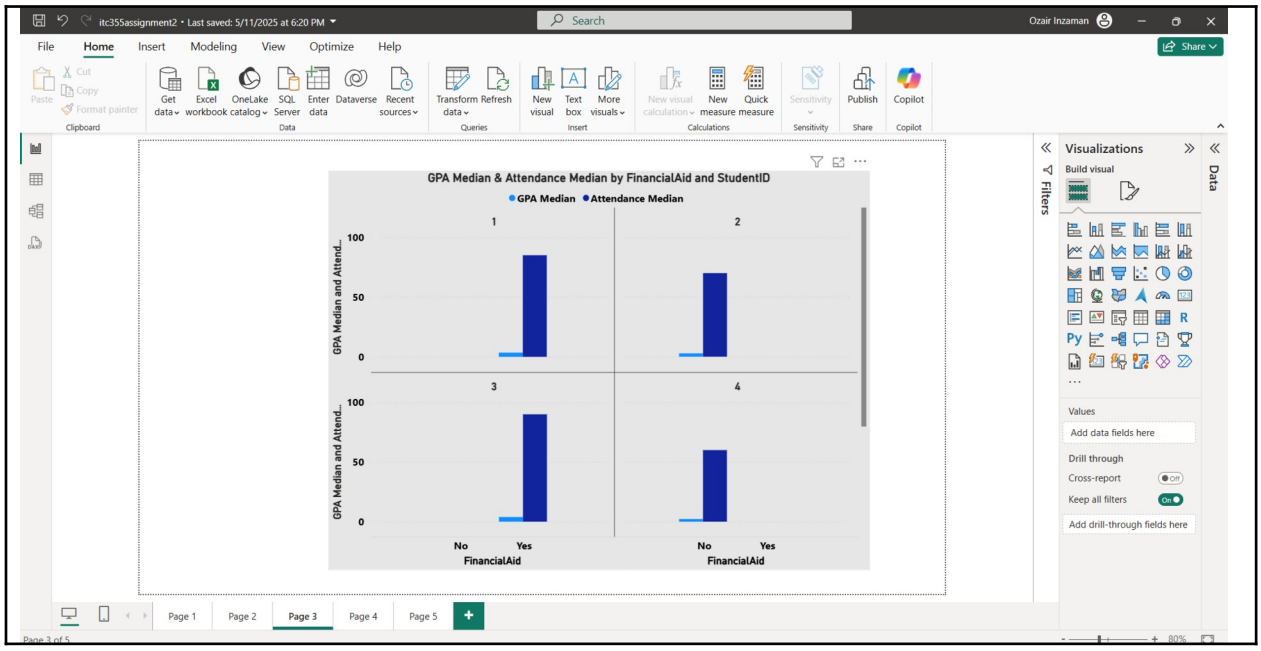
Predictive Analytics was used to predict future events using historical data such as those based on past behavior and identifying patterns.. Predictions based on students that are likely to drop out were made using the **GPA**, **attendance**, financial aid. and **mental health status**.

Graph 2: Impact Of Attendance Based On GPA



The graph above shows how attendance has an impact on the GPA based on count. Students with a low attendance & GPA are always at a higher risk. Attendance is a major factor of engagement and retention. Poor attendance is interrelated with high dropout rates. Analyzing attendance against the dropout status will assist to identify students who are disengaged & face a higher risk of leaving the university.

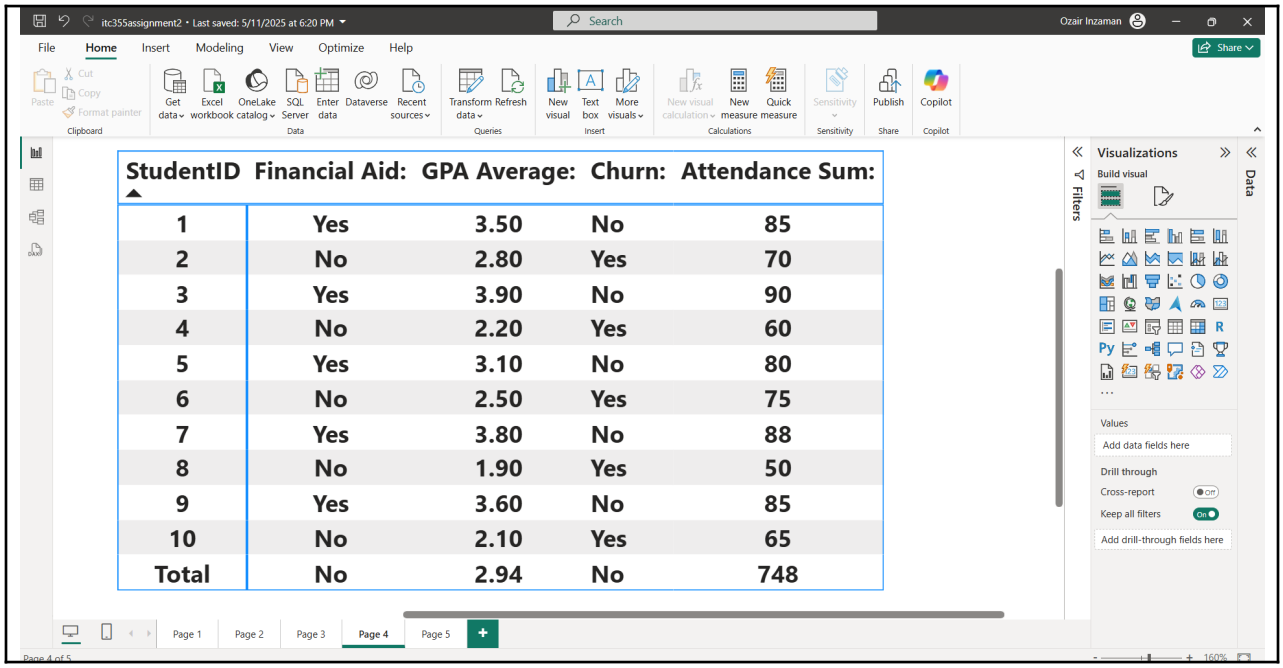
Graph 3: Relation Between Financial Aid, GPA Median & Students



This bar chart above illustrates the relationship between financial aid, GPA median and students. Students with a high gpa and have received financial aid are less likely to drop out while those with a less gpa and no financial aid are more likely to drop out.

Financial aid is a huge factor that determines the success rate of students. A student with less gpa and no financial aid will face a negative impact thus increasing the risk of student dropouts while those with a high gpa with received financial aid are less likely to dropout. Analyzing this information will assist in identifying students who require the necessary assistance.

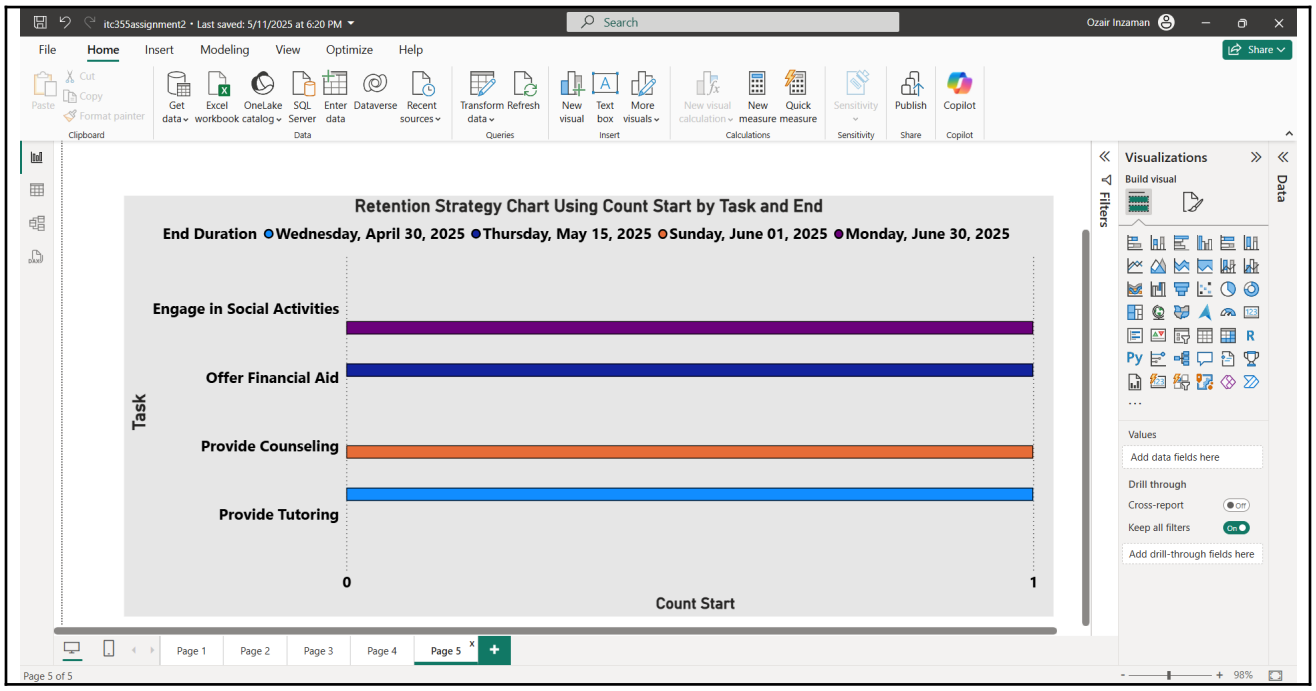
Graph 4: Financial Aid On Student Retention



StudentID	Financial Aid:	GPA Average:	Churn:	Attendance Sum:
1	Yes	3.50	No	85
2	No	2.80	Yes	70
3	Yes	3.90	No	90
4	No	2.20	Yes	60
5	Yes	3.10	No	80
6	No	2.50	Yes	75
7	Yes	3.80	No	88
8	No	1.90	Yes	50
9	Yes	3.60	No	85
10	No	2.10	Yes	65
Total	No	2.94	No	748

This chart above shows the outcome of financial aid on retention. If students with no financial aid have a higher dropout rate, then financial support programs could be implemented to reduce attrition levels. Financial aid is the main factor in student retention. Students who don't get any form of financial aid do struggle with school fees therefore this increases the dropout rates.

Graph 5: Retention Strategy Chart



The chart above outlines key tasks which are **Provide Tutoring**, **Offer Financial Aid**, **Provide Counseling** & **Engage in Social Activities** with each task containing the end duration as stated below the chart title. This is aimed to improve student retention & reduce the dropout rates.

The tasks have been color-coded to differentiate the various retention strategies. This will assist the university to keep track of the timing & be aware of the various retention strategies thus making it easy to allocate resources and monitor the effectiveness of these efforts.

3. BI Solution Design & Dataset:

3.1 Data Processing:

The data was imported using the csv files in Microsoft Power BI Desktop which are **churn_analysis.csv, feature_importance.csv, retention-strategy.csv & student_performance.csv.** The **retention-strategy.csv** is an additional csv file created by referring to the R algorithm from assignment 1. The data was extracted from the R algorithm then converted to a CSV file then imported and interpreted on the Power BI tool.

Retention Strategy Algorithm:

```
# Sample data for Gantt chart
gantt_data <- data.frame(
Task = c("Provide Tutoring", "Offer Financial Aid", "Provide Counseling", "Engage in Social Activities"),
Start = as.Date(c("2025-04-01", "2025-04-15", "2025-05-01", "2025-06-01")),
End = as.Date(c("2025-04-30", "2025-05-15", "2025-06-01", "2025-06-30"))
)
# Plot Gantt chart
ggplot(gantt_data, aes(x = Start, xend = End, y = Task, yend = Task)) +
geom_segment(size = 8, color = "skyblue") +
labs(title = "Retention Strategy Timeline", x = "Date", y = "Task") +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

3.2 Datasets Used For Analysis in CSV Format

Below are the screenshots containing the csv data used to generate the various visual charts for interpretation of data.

churn_analysis.csv:

StudentID	Attendance	FinancialAid	Dropout
1	85	Yes	No
2	70	No	Yes
3	90	Yes	No
4	60	No	Yes
5	80	Yes	No
6	75	No	Yes
7	88	Yes	No
8	50	No	Yes
9	85	Yes	No
10	65	No	Yes

feature_importance.csv:

StudentID	GPA	Attendance	FinancialAid	Churn
1	3.5	85	Yes	No
2	2.8	70	No	Yes
3	3.9	90	Yes	No
4	2.2	60	No	Yes
5	3.1	80	Yes	No
6	2.5	75	No	Yes
7	3.8	88	Yes	No
8	1.9	50	No	Yes
9	3.6	85	Yes	No
10	2.1	65	No	Yes

retention-strategy.csv:

Task	Start	End
Provide Tutoring	2025-04-01	2025-04-30
Offer Financial Aid	2025-04-15	2025-05-15
Provide Counseling	2025-05-01	2025-06-01
Engage in Social Activities	2025-06-01	2025-06-30

student_performance.csv:

StudentID	GPA	Attendance	FinancialAid	MentalHealthStatus
1	3.5	85	Yes	Low
2	2.8	70	No	Medium
3	3.9	90	Yes	Low
4	2.2	60	No	High
5	3.1	80	Yes	Medium
6	2.5	75	No	Low
7	3.8	88	Yes	Medium
8	1.9	50	No	High
9	3.6	85	Yes	Low
10	2.1	65	No	High

Note: The retention strategy data was converted from the R algorithm into a csv format for a suitable representation of data in a visualization format.

4. Results, Interpretation & Evaluation:

Firstly, the **descriptive analysis** provided insights on issues that influenced student dropouts which are a low GPA, poor attendance, mental health challenges & financial problems. Visualizing these issues using the Power BI tool will allow the university administrators to gain a better understanding on where to prioritize their retention efforts.

Secondly, the **predictive model** shall make use of these insights in order to identify those students who are at a high risk of dropping out then take the suitable steps such as providing additional academic assistance, targeted tutoring, financial aid, or mental health counseling.

Thirdly, universities can make use of the **Power BI's visuals** & **used data** to keep track of high risk students to assign resources effectively thus reducing dropout rates.The **GPA** and **attendance** have been the most significant predictors of student dropout.

5. Discussion, Recommendations, and Future Work:

Descriptive & predictive analytics was used with the Microsoft Power Bi Desktop tool in order to solve the problem of student dropouts. Students facing a high risk of dropping out were identified considering the GPA level and receival of financial assistance then strategies such as additional academic assistance, targeted tutoring, financial aid and others were implemented.

In the future, gradient boosting or neural networks can be used to enhance the accuracy of prediction. Also extra features such as social engagement can be added to improve the accuracy in a chart.

6. References

Fernan (2020). *How to create BAR CHARTS in Power BI // Beginners Guide to Power BI in 2020*. [online] YouTube. Available at: https://www.youtube.com/watch?v=_LYN02Fi4Wo [Accessed 9 May 2025].

www.youtube.com. (n.d.). *How to Export Data From R to a CSV File | Basics of R Programming*. [online] Available at: https://www.youtube.com/watch?v=D9-at2sPQgs.

Pavan Lalwani (2020). *4.5 How to Create a Matrix in Power BI | Power BI Tutorials for Beginners | By Pavan Lalwani*. [online] YouTube. Available at: https://www.youtube.com/watch?v=hCrxEzuHAmk [Accessed 9 May 2025].